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Antioxidant-Rich Fenugreek Seed Oil: Potential Applications in Food Preservation and Health Promotion

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Abstract

The escalating demand for clean-label, plant-derived antioxidants necessitates evaluation of underutilized regional resources. This study characterized fenugreek (*Trigonella foenum-graecum*) seed oil from Sokoto, Nigeria, for nutraceutical applications. Soxhlet extraction yielded 13.56% oil containing bioactive phytochemicals (flavonoids, saponins, steroids, terpenoids, tannins). Antioxidant evaluation demonstrated strong DPPH radical scavenging (66.25% mean inhibition) and moderate phenolic content (30.445 mg GAE/100g). Concentration-dependent activity (0.3–0.5 mg/mL) showed excellent linear correlation ($r = 0.999$, $p < 0.01$), indicating reliable dose-response. These properties position fenugreek seed oil as a viable natural antioxidant for food systems and dietary supplements, with potential economic benefits for Nigerian agriculture.

Plant-derived anti-oxidants offer multifunctional benefits: antioxidant protection, antimicrobial activity, and nutritional enhancement

Keywords: *Natural antioxidants, fenugreek, functional ingredients, DPPH, phenolic compounds, clean label*

1. **The Natural Antioxidant Imperative**

Synthetic antioxidants (BHA, BHT) face mounting regulatory and consumer pressure due to safety concerns. Plant-derived alternatives offer multifunctional benefits: antioxidant protection, antimicrobial activity, and nutritional enhancement (Shahidi & Ambigaipalan, 2015). Fenugreek, a GRAS-listed herb with established culinary use, presents an attractive candidate for natural antioxidant applications.

2. Materials and Methods

Processing:

Seeds are subjected to drying (40°C) then to milling then extracted with Soxhlet extraction (n-hexane, 3h) then concentrated using rotary evaporation then to storage (4°C)

Quality Assessment:

a. Phytochemical screening (standard colorimetric methods)
b. DPPH radical scavenging (517 nm, ascorbic acid reference)
c. Total Phenolic Content (Folin-Ciocalteu, gallic acid standard)

3. Findings

Parameter	Finding	Implication
Extraction yield	13.56%	Commercial viability
Phytochemical diversity	5/6 classes detected	Multiple bioactivities
DPPH IC_{50} trend	59.95–72.30% (0.3–0.5 mg/mL)	Effective at low concentrations
TPC	30.445 mg GAE/100g	Moderate-high phenolic contribution
Linearity (r^2)	0.998	Predictable performance

4. Food Application Potential

Oil Systems: Phenolic compounds inhibit lipid peroxidation through radical scavenging and metal chelation.

Meat Products: Saponins and tannins may extend shelf-life while reducing synthetic additive requirements.

Bakery: Flavonoid stabilization of fats during high-temperature processing.

5. Economic and Agricultural Considerations

Fenugreek cultivation in Northern Nigeria offers:

a. Drought tolerance suitable for Sahelian conditions
b. Nitrogen-fixation improving soil fertility
c. Multiple harvestable products (seeds, leaves, fodder)
d. Export potential for value-added extracts

6. Recommendations

a. Stability trials in model food systems
b. Sensory evaluation of fortified products
c. Scale-up extraction optimization
d. Farmer training on quality cultivation practices

10. Transparency Statement

No conflicts of interest were identified that could compromise scholarly objectivity.

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